

Learning Objectives

- Explain what AI is.
- Explain how generative AI is positioned within AI technologies.

When you use a phone, AI is often at work. A spam filter sorts your email, a store recommends products you might buy, a translation app changes one language into another, and ChatGPT generates text from a prompt. These are all examples of AI, but they are not all the same kind. AI includes machine learning, within machine learning is deep learning, and today's generative AI systems are built on deep learning — a series of nested technologies, each more specialized than the last.

? Guiding question: What is AI, and how are machine learning, deep learning, and generative AI related?

✎ Explore · In Pairs

With a partner, list three tasks a phone or computer does for you that seem to need "intelligence" (for example, sorting spam, recommending a video, or translating text). Then predict: does the computer follow fixed rules, or does it learn from examples? Give one reason.

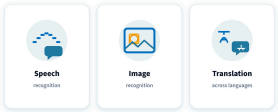
📌 Point!

1. What is Artificial Intelligence (AI)?

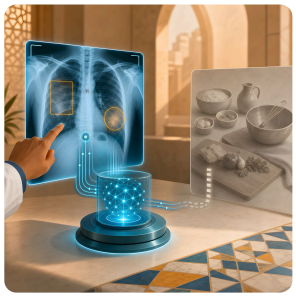
(1) **(AI (Artificial Intelligence))**…a general term for technologies that reproduce or perform intelligent human behavior (learning, reasoning, judgment, etc.) on a computer.
e.g., speech recognition, image recognition, translation

2. The Relationship between AI and Machine Learning

(1) Within AI technologies, there is machine learning; within machine learning, there is deep learning; and within deep learning, there is generative AI. The broad field of AI includes increasingly specialized technologies as a series of nested categories.



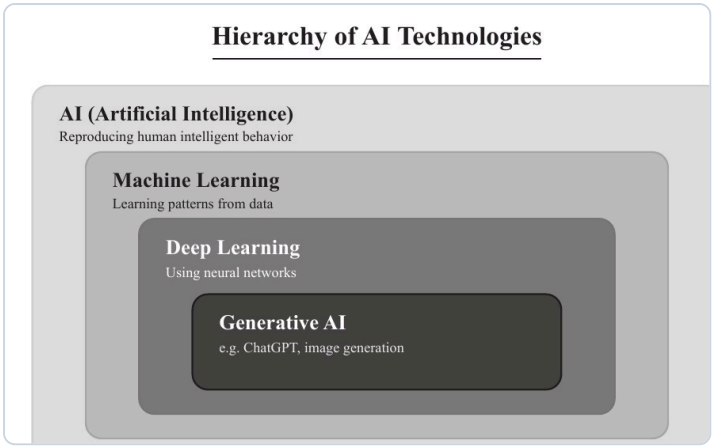
Three everyday AI examples: speech, image, translation.



Today's AI is narrow — expert at one task only.

💡 THINK IT THROUGH

Spam filters and product recommendations are both examples of machine learning, but the tasks they perform are completely different. From the perspective of how AI works, explain what these two applications have in common.

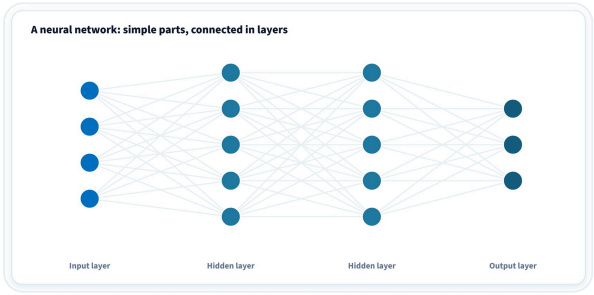


Hierarchy of AI Technologies

(2) **(Machine Learning)**…one of the learning technologies that makes AI work. It learns patterns from data to make predictions and judgments.
e.g., spam filters, product recommendations

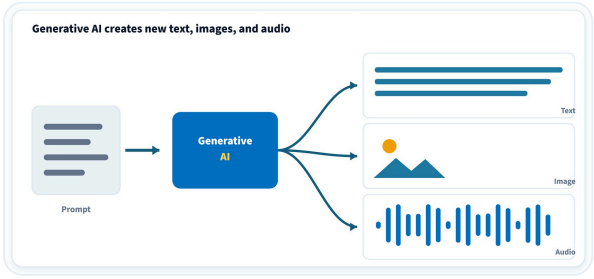
(3) **(Deep Learning)**…an advanced technology within machine learning that uses neural networks. It learns complex patterns using large-scale data.
e.g., image analysis for autonomous driving, speech synthesis

(4) **(Neural network)**…a system modeled after the workings of the nerve cells (neurons) of the human brain. By connecting many components, it learns from data and becomes capable of making complex judgments. It is the core technology supporting recent advances in AI.

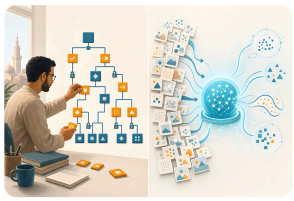


A neural network connects many simple components.

(5) **(Generative AI)**…AI technology that uses deep learning to generate new data (text, images, audio, programs, etc.).
e.g., ChatGPT, image generation AIs



Generative AI creates new text, images, and audio.



Fixed hand-written rules versus rules learned from data.

PAUSE & THINK

Deep learning needs large-scale data to learn. Why might an AI struggle with something it has rarely seen in its data?

KEY TERMS

neural network — a system modeled after the nerve cells (neurons) of the human brain, used to learn from data and make complex judgments.

KEY TERMS

generative AI — AI that uses deep learning to generate new data such as text, images, and audio.

✎ Worked Example

(1) Mark each statement (A–D) with ○ if true or × if false.

A Machine learning is one of the technologies that makes AI work.

B Deep learning is a completely different technology from machine learning.

C Generative AI is a technology that uses deep learning to generate new data.

D AI and machine learning have the same meaning.

(2) Match each description (a–c) with the most appropriate term from the choices below (A–C).

a A technology that learns patterns from data to make predictions and judgments

b A technology that uses neural networks to learn complex patterns

c AI technology that generates new data such as text and images

[Choices] **A** Machine learning **B** Deep learning **C** Generative AI

✓ Solution

(1)

A Machine learning is one of the learning technologies that makes AI work. Therefore, ○.

B Deep learning is a type of machine learning (an advanced technology within machine learning). Therefore, ×.

C Correct description of generative AI. Therefore, ○.

D AI is a broad concept, and machine learning is one of the technologies within it. They do not have the same meaning. Therefore, ×.

(2)

a: Learning patterns from data is machine learning. **A**

b: Using neural networks is deep learning. **B**

c: Generating new data is generative AI. **C**

KEY TERMS

machine learning — learns patterns from data.

deep learning — uses neural networks on large-scale data.

Try

1 Answer the following questions.

1. What is the general term, expressed by a two-letter abbreviation, for technologies that reproduce or perform intelligent human behavior on a computer?
2. What is the term for the learning technology that makes AI work by learning patterns from data to make predictions and judgments?
3. What is the term for the advanced technology within machine learning that uses neural networks?
4. What is the term for the AI technology that uses deep learning to generate new data?

2 Answer the following questions.

(1) From the following options (A–D), choose the one that correctly describes the relationship between AI, machine learning, deep learning, and generative AI.

A AI > Deep Learning > Machine Learning > Generative AI

B Machine Learning > AI > Generative AI > Deep Learning

C AI > Machine Learning > Deep Learning > Generative AI

D Generative AI > Deep Learning > Machine Learning > AI

(2) From the following options (A–D), choose the one that is NOT an appropriate example of generative AI.

A ChatGPT

B Image generation AI

C Spam filter

D Audio generation AI

Think as an Engineer — Investigate & Decide

Starting from the Think It Through note about hallucination, investigate further and decide.

1 · Investigate. Find one documented example where a generative AI gave an answer that sounded correct but was actually wrong (a "hallucination"). Record its source — a news article, an official report, or a prompt you tested yourself with a screenshot — and note where you found it. Do not rely on an example recalled from memory without a source.

2 · Weigh it up. Note one benefit and one risk of using generative AI to write a school report — for the student, and for the teacher.

3 · Decide. Should generative AI be allowed for homework in your school? Recommend one rule and give two reasons.

EXAM-STYLE QUESTION

Analyze the relationship between AI, machine learning, deep learning, and generative AI. **[6] In your answer:** use the idea of nested categories, and give one example of each.

THINK IT THROUGH

Generative AI can produce text that sounds plausible but is factually incorrect (a "hallucination"). Explain why it is dangerous to use such output as the answer to a school report as-is, and state what learners should do instead.

Stuck? One benefit is speed; one risk is that the answer may be factually incorrect, so it must be checked.

🌍 In a New Context

A farmer wants to use AI to distinguish healthy crops from diseased ones in photographs. Using what you learned about deep learning and image analysis, explain what the AI would need to learn from, and identify one reason its judgment might be wrong.

❓ Lesson Question — Answered

The lesson question asked what AI is and how machine learning, deep learning, and generative AI are related. AI is a general term for technologies that reproduce or perform intelligent human behavior on a computer, and machine learning, deep learning, and generative AI are increasingly specialized fields nested within it rather than parallel to it. The key idea from this lesson is that these are not four separate technologies to learn in isolation, but one broad field understood at different levels of specialization.

KEY TERMS

deep learning — uses neural networks to learn complex patterns.

generative AI — uses deep learning to generate new data.

Exercise

1 Cover the Point! section with a red plastic sheet and quiz yourself in your notebook in numerical order.

2 Read the following passage and answer each question.

The general term for technologies that reproduce or perform intelligent human behavior on a computer is (a). One of the learning technologies that makes (a) work by learning patterns from data is (b). Within (b), an advanced technology that uses neural networks is (c). Furthermore, the technology that uses (c) to generate new data is (d).

1. Fill in the blanks (a)–(d).

⭐ REFLECT & CHALLENGE

Reflect: Before this lesson, did you think AI was one single technology? How has your idea changed? Which of the four — AI, machine learning, deep learning, generative AI — was hardest to tell apart, and why?

Challenge: Generative AI can create text, images, and audio. Suggest one helpful use and one possible misuse, and say how the misuse could be reduced.

3 Answer the following questions.

(1) For each example (a–d), indicate which of the following is most closely related: **A** Machine learning, **B** Deep learning, **C** Generative AI.

a Automatically sorting spam emails

b Recognizing pedestrians on the road in autonomous driving

c Automatically generating new images from text

d Recommending products based on a user's purchase history

(2) From the following options (A–D), choose the one that most appropriately describes the relationship between AI, machine learning, deep learning, and generative AI.

A Machine learning is a type of AI, and deep learning is a type of machine learning.

B AI is a type of machine learning, and generative AI is a type of AI.

C Deep learning and machine learning are completely different technologies.

D Generative AI is a technology that generates data without using machine learning.

⭐ Key takeaway

Remember:
AI is the broad field; machine learning sits inside it, deep learning inside machine learning, and today's generative AI is built on deep learning — from the broadest idea to the most specialized.